

## Software Defined Networking (SDN) – Brief Explanation

- **Software Defined Networking (SDN)** is a networking architecture where the **control plane** (decision-making) is separated from the **data plane** (packet forwarding).
- In SDN, a **centralized software controller** manages and controls the network instead of each router making its own decisions.
- The **SDN controller** calculates routes, applies network policies, and programs network devices like **switches and routers**.
- This centralized control allows **easier network management, better flexibility, and faster configuration** compared to traditional networks.

## Control Plane vs Data Plane (Brief)

### Control Plane

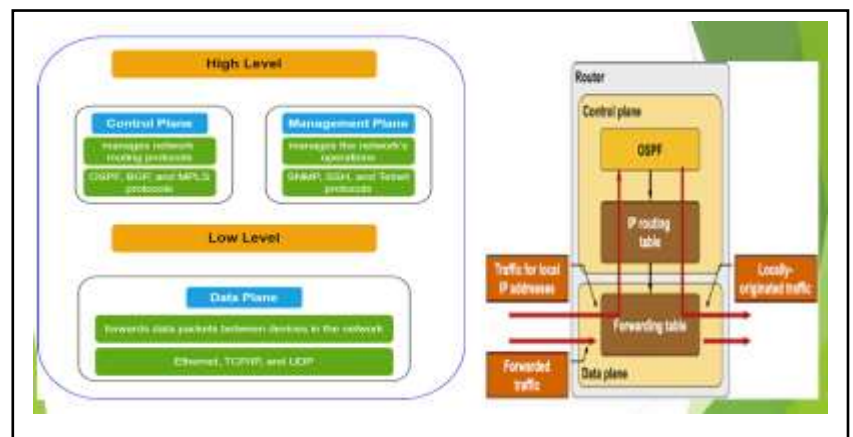
- Determines **how network packets are routed and processed**.
- Acts as the **brain of the network**.
- Handles **signaling and routing messages**.
- **Builds routing tables and network policies**.

Control plane builds routing tables and policies, whereas the data plan operates based on these rules to move traffic efficiently.

### Data Plane

- **Forwards packets** to their destinations.
- Acts as the **muscle of the network**.
- Carries the **actual payload/data traffic**.
- **Operates using the rules** created by the control plane.

## Brief Explanation of the Figure



- **High Level (Control & Management Planes):**
  - **Control Plane:** Manages **network routing decisions**. It uses routing protocols like **OSPF, BGP, and MPLS** to build the **IP routing table**.
  - **Management Plane:** Handles **network administration and monitoring** using protocols like **SNMP, SSH, and Telnet**.
  
- **Low Level (Data Plane):**
  - **Data Plane:** Responsible for **forwarding data packets between devices** in the network.
  - It uses protocols like **Ethernet, TCP/IP, and UDP** to move traffic based on the **forwarding table**.
  
- **Router Operation (Right Side of Figure):**
  - The **control plane** calculates routes and updates the **IP routing table**.
  - The **data plane** uses this information to create a **forwarding table** and **send packets to the correct destination**.
  - Traffic can be **locally generated** by the router or **forwarded** to other networks.

### Router Role (Brief)

- **Routing protocols** (e.g., OSPF, BGP) **build the routing table**.
- ★ • The router **selects the best routes** from the routing table.
- The **best routes are copied to the forwarding table**.
- The **data plane uses the forwarding table to forward packets** to the correct destination.

## Forwarding Table (Meaning)

- A **forwarding table** is a table inside a router or switch that **tells the device where to send incoming packets**.

## Traditional Networking vs SDN (Brief)

### Traditional Networking

- **Routers make routing and forwarding decisions** themselves.
- In traditional networks, **routers will take** the decisions of routing and forwarding the data
- Decisions are based on **routing algorithms and routing tables**.
- Routers contain both:
  - **Control Plane** – makes routing decisions.
  - **Data Plane** – forwards packets.
- Common routing protocols: **OSPF, RIP, BGP**.

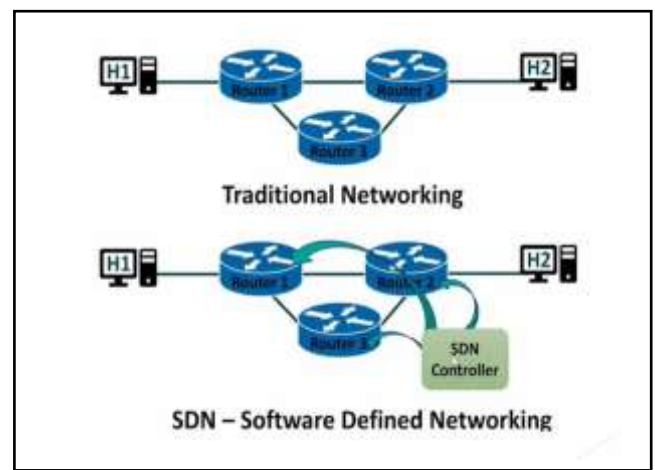
### Software Defined Networking (SDN)

- **Routers do not make routing decisions**.
- A **central SDN controller** decides how traffic should be routed.
- The controller **sends instructions to routers/switches** on where to forward data.
- **Network control is managed through software**, making the network more flexible and easier to manage.



In SDN, routers will not take decisions, SDN controller will give the command to router where to forward the data. Software inside SDN controller will do decisions

SDN Controller - Central software that controls the entire network.



Upper Figure

- Shows the difference between **Traditional Networking** and **SDN (Software Defined Networking)**.
- **Traditional Networking:**
  - Each **router (Router 1, 2, 3)** makes its **own routing decisions**.
  - Control and forwarding are handled **inside each router**.
- **SDN Networking:**
  - A **central SDN Controller** manages the network.
  - Routers/switches **do not decide routes themselves**.
  - The **controller tells the routers where to forward traffic**.

Lower Figure

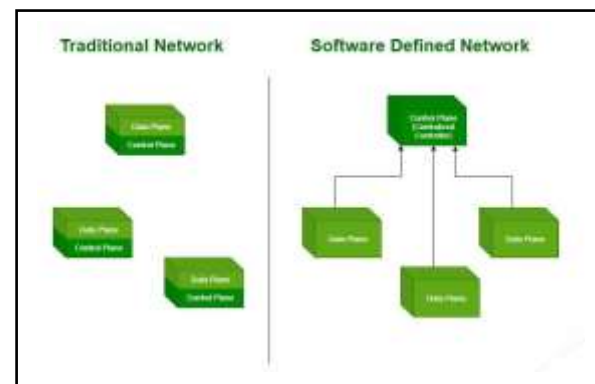
- Shows how **control plane** and **data plane** are organized.

#### Traditional Network

- Every device has **both Control Plane and Data Plane**.
- Each router independently **decides routes and forwards packets**.

#### Software Defined Network

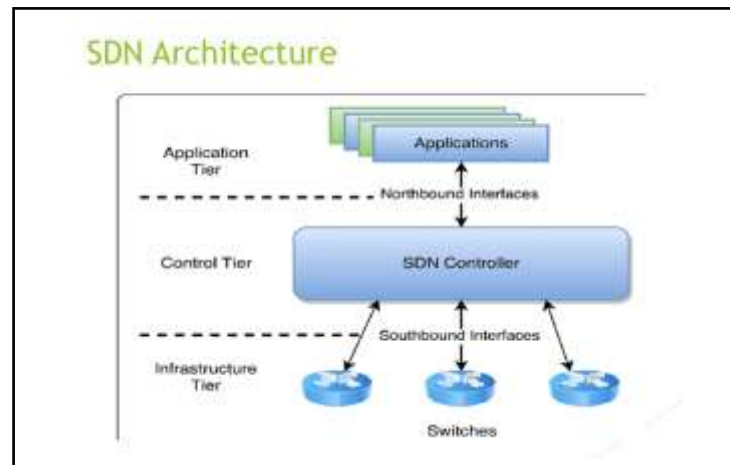
- The **Control Plane is centralized** in the **SDN Controller**.
- Network devices only contain the **Data Plane**.
- The controller **controls and programs all devices centrally**.



### Main Idea of the Slide:

SDN **separates control from forwarding** and **centralizes network control**, while traditional networking keeps **both functions inside each router**.

## SDN Architecture



This diagram shows the **three layers (tiers) of Software Defined Networking**.

- **Application Tier**
  - Contains **network applications** (security, traffic management, monitoring).
  - Communicates with the controller through **Northbound Interfaces/APIs**.
- **Control Tier**
  - Contains the **SDN Controller**.
  - It acts as the **brain of the network** and decides routing, policies, and traffic flow.
- **Infrastructure Tier**
  - Contains **network devices like switches/routers**.
  - These devices **forward data packets** based on instructions from the controller.
- **Southbound Interfaces**
  - Used for communication between the **SDN Controller and network devices**.

### Main idea:

Applications → talk to → **SDN Controller** → which controls → **network devices**.

## SDN Components



## SDN Components

This diagram shows the **main components of SDN architecture**.

- **SDN Applications** – Programs that manage network services (security, load balancing, monitoring).
- **SDN Controller** – Central software that **controls the entire network**.
- **Northbound APIs** – Connect **applications to the SDN controller**.
- **Southbound APIs** – Connect **controller to network devices**.
- **Network Devices** – Switches/routers that **forward packets**.

### Main idea:

The SDN network works using **applications, controller, APIs, and network devices** that communicate to manage the network centrally.

## Comparison between Software-Defined Networking (SDN) and Traditional Networking

### 1. Architecture

- **SDN:** Separates the control plane from the data plane.
- **Traditional Networking:** Control plane and data plane are combined in the same device.

### 2. Nature

- **SDN:** Software-centric (controlled mainly by software).
- **Traditional:** Hardware-centric (depends on physical networking devices).

### 3. Control Plane

- **SDN:** Managed by a centralized controller.
- **Traditional:** Distributed across each networking device.

### 4. Configuration

- **SDN:** Automatic and centrally managed configuration.
- **Traditional:** Manual configuration for each device.

### 5. Deployment Complexity

- **SDN:** Faster setup and easier to adapt.
- **Traditional:** Takes more time and is less flexible.

### 6. Programmability

- **SDN:** Programmable through software.
- **Traditional:** Not easily programmable.

### 7. Scalability

- **SDN:** Highly scalable for growing networks.
- **Traditional:** Limited scalability.

### 8. Cost

- **SDN:** Lower maintenance cost and more cost-effective.
- **Traditional:** Higher maintenance cost.

## 9. Flexibility

- **SDN:** Dynamic and adaptable.
- **Traditional:** Static and siloed.

✅ In short: **SDN is more flexible, programmable, and centrally controlled**, while **traditional networking relies on hardware and manual configuration**.

## Advantages of SDN (Software Defined Networking)

### 1. Control

- Network managers can **control traffic flow easily**.
- **Switch rules can be changed based on network needs**.
- Supports **flexible multi-tenant cloud environments**.

### 2. Efficiency

- **Virtualized network services** reduce the need for expensive hardware.
- Improves **resource utilization and network performance**. (optional)

### 3. Management

- Network can be **managed from a centralized SDN controller**.
- Administrators can **apply policies to many switches at once** without configuring each device separately.

### 4. Visibility

- The **central controller monitors network traffic**.
- It can **detect suspicious activity and redirect packets** to improve security and performance.



## Common use cases of Software-Defined Networking (SDN) and their benefits

### ◆ 1. Data Center Optimization

👉 Used inside data centers (servers, storage, networking)

#### Benefits:

- ✓ Resources (CPU, storage) are **allocated automatically**
- ✓ Systems can **scale up/down automatically**
- ✓ Network traffic is **optimized (no congestion)**

Simple idea:

**Efficient and automated data center management**

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### ◆ 2. Cloud Networking

👉 Used in cloud systems (AWS, Azure, etc.)

#### Benefits:

- ✓ On-demand services (use anytime)
- ✓ Supports **multi-cloud & hybrid cloud**
- ✓ Better **bandwidth control & network slicing**

Simple idea:

**Flexible and scalable cloud communication**

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### ◆ 3. IoT Networking

👉 Used in IoT systems (sensors, ESP32, smart devices)

#### Benefits:

- ✓ Manages many IoT devices
- ✓ Prioritizes **important data (e.g., alerts)**
- ✓ Adapts to network changes dynamically

Simple idea:

**Smart handling of IoT device communication**

Three layers of an IoT (Internet of Things) system.

### 1. Perception Layer

- The **lowest layer** of IoT.
- Contains **sensors and actuators** that collect data from the physical environment.
- **Examples:** temperature sensors, motion sensors, smart lights, water sensors.

### 2. Network Layer

- Responsible for **transmitting data** from sensors to servers or cloud systems.
- Uses **routers, gateways, and communication networks** like Wi-Fi, Bluetooth, or cellular networks.

### 3. Application Layer

- The **top layer** where data is processed and used by applications.
- Provides **services to users** through cloud platforms, servers, or mobile apps.

**Real-Life Example:** **Smart Home Temperature System**

- **Perception Layer:** Temperature sensor detects room temperature.
- **Network Layer:** Wi-Fi router sends the data to the cloud server.
- **Application Layer:** A mobile app shows the temperature and allows the user to control the air conditioner.

**In short:** IoT works by **collecting data (Perception)**, **sending it through the network (Network)**, and **providing useful services to users (Application)**.

